Connecting Drone Intelligence to Real-Time Decision-Making

Problem Provider: ArcDrone
Theme: Defense & Security

1. The Problem

Drones can use computer vision to detect people, vehicles, and other objects on the ground, but those detections are only useful if they reach an operator in time to respond.

Without a reliable wireless connection between the drone and its ground control station, important detections may be missed or arrive too late.

Existing approaches often rely on operators continuously watching live video or drones operating autonomously without a clear way for a person to intervene when something unexpected happens.

2. Background & Context

Autonomous drones can process camera footage onboard and identify objects using computer vision. However, detecting an object is only one part of the process. The drone must also transmit that information to ground control software, where an operator can review it and take action.

Rather than transmitting continuous video, a drone could send a smaller message containing the information needed to identify and respond to a detection.

Solutions should consider:

- **Hardware and software integration:** Establish a working wireless connection between the drone side and ground station.
- **Speed:** Deliver detection alerts quickly enough for an operator to respond.
- **Reliability:** Account for dropped messages or temporary connection loss.
- **Accessible hardware:** Use off-the-shelf wireless components that can be configured during the hackathon.
- **Human control:** Include a clear way for an operator to respond to an alert or initiate a simulated override.

3. Your Challenge

How might we build a reliable wireless communication system that sends drone detections to a ground control station and allows an operator to respond?

Teams should develop an end-to-end system that identifies an object using computer vision, transmits an alert through a physical wireless connection, and delivers that information to ground control software.

The focus is on connecting detection to operator action, rather than developing a new computer vision model.

4. Potential Project Directions

Teams may consider exploring:

- **Object detection:** Use an existing computer vision model to identify people, vehicles, or other objects in sample drone footage.
- **Wireless communication:** Transmit detection alerts using hardware such as LoRa, RF modules, or ESP-NOW.
- **Alert design:** Determine what information to transmit, such as object type, detection confidence, timestamp, or location within the image.
- **Reliable transmission:** Implement message acknowledgements, retries, or connection-loss handling.
- **Ground control interface:** Display incoming detections and give operators a clear way to respond.
- **Operator override:** Demonstrate a simulated command sent from the ground station back to the drone.

These are suggestions, not requirements. Teams are encouraged to develop their own approaches.

Solutions must include a working wireless hardware connection, not just communication simulated within one computer. Teams should use public sample data rather than ArcDrone's live systems, customer sites, or proprietary models.

5. Skills & Areas of Interest

This challenge may be particularly interesting to participants with experience or interest in:

- Embedded systems and microcontrollers
- Wireless communication and networking
- Computer vision and object detection
- Hardware and software integration
- Python or other programming languages
- UX/UI and ground control software
- Robotics and autonomous systems

Teams can use existing computer vision models. Training a new model is not required.

Additional resources:

- VisDrone dataset and Ultralytics integration – Drone imagery for object detection.
- Community-trained VisDrone models are available on Hugging Face.

6. What Success Could Look Like

Teams should focus on demonstrating one complete detection-to-response process using sample footage and a working wireless connection.

A prototype might:

- Detect an object in sample drone footage using an existing model.
- Send a compact detection message over a physical wireless link.
- Display the received alert in a ground control interface.
- Handle a dropped message or temporary connection loss.
- Allow an operator to acknowledge an alert or trigger a simulated override command.

The goal is to demonstrate a reliable connection between what a drone detects and what an operator can do in response, without requiring a fully operational drone or production-ready control system.